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Publication list (last updated July 2026)

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The publication list including the citation numbers is created in NASA/ADS (here).

First Author

6. **Y. Nakazato**, K. Matsumoto, A.K. Inoue, et al., *Clump-Scale Dust Attenuation in Epoch of Reionization Galaxies: Spatially Resolved Properties from FirstLight Simulations* , 2026, accepted to ApJ, arXiv:2602.07347
   - *We computed spatially resolved attenuation curves using zoom-in simulations and post-processing dust radiative transfer.*
5. **Y. Nakazato**, S. Kazuyuki, A.K. Inoue, M. Ricotti, *Unveiling the Ionized and Neutral ISM at $z > 10$: The Origin of [OIII]/[CII] Ratios from a Sub-parsec Resolution Radiative Transfer Simulation*, 2026, ApJ, 998, 1, 26
   - *We computed [OIII]88μm and [CII] 158μm luminosities for simulated galaxies at $z > 10$, and investigated the physical origins of [OIII]/[CII] ratio enhancement at the high-z Universe.*
4. **Y. Nakazato**, A. Ferrara, *Radiatively driven dusty outflows from compact galaxies at $z > 10$*, MNRAS, 544, 4, 2025
   - *We calculated the modified Eddington luminosity considering dusty gas and showed radiation-driven outflow could explain the UV excess at $z > 10$.*
3. **Y. Nakazato**, D. Ceverino, N. Yoshida, *A merger-driven scenario for clumpy galaxy formation in the epoch of reionization: Physical properties of clumps in the FirstLight simulation*, ApJ, 975, 2, 238, 2024
   - *We showed that observable 100 pc scale clumps at $z > 6$ are mainly formed by tidal-tails of major mergers.*
2. **Y. Nakazato**, N. Yoshida, D. Ceverino, *Simulatioins of high-redshift [OIII] emitters: Chemical evolution and multi-line diagnostics*, ApJ, 953, 14, 2023
   - *We devised an emission model of HII regions in cosmological zoom-in simulations and investigated [OIII]-emitters at $z = 6 - 9$.*
1. **Y. Nakazato**, G. Chiaki, N. Yoshida, et al., *H_2 cooling and gravitational collapse of supersonically Induced gas objects*, ApJL, 927, 1, 2022
   - *We performed cosmological simulations and showed the formation of dark matter-less objects via baryon-dark matter relative velocities at $z \sim 20$.*

First-Authored Proceedings & Other Reports (Refereed)

3. **Y. Nakazato**, Kosei Matsumoto, Tohru Nagao, et al., *Rapid Carbonaceous Dust Evolution in EoR Galaxies: Synergies Among JWST, PRIMA, and ALMA*, PRIMA General Observer Science Book Volume 2
2. **Y. Nakazato**, *A merger-driven scenario for clumpy galaxy formation in the epoch of reionization: Physical propertiesof clumps in the FirstLight simulation*, Proceedings of International Astronomical Union, Volume 391

1. **Y. Nakazato**, G. Chiaki, N. Yoshida, et al., *The formation of Supersonically Induced Gas Objects (SIGOs) with H_2 cooling*, Proceedings of International Astronomical Union, Volume 362, 2023

Co-Author

summary: **total 29 papers/ (∗) 6 papers as the 2nd or 3rd author**

†: Responsible for theoretical interpretation in an observational paper

¶: Generated realistic mock images for an observational paper

29. M. Xiao, et al. (incl. **Y. Nakazato**), *A Census of the 200 Most Massive Galaxies Spectroscopically Observed with JWST at $z_{\rm spec} \sim 3 - 15$*, 2026, submitted to A&A, arXiv:2606.30802

*28. D. Toyouchi, A. Ferrara, **Y. Nakazato**, K. Matsumoto, R. Schneider, K. Otaki, *Grain-size evolution and rapid dust growth in high-redshift galaxies*, 2026, submitted to MNRAS, arXiv:2606.06108

†27. J. Song, M. Ouchi, T. Kiyota, C. Zhu, X. Kong, **Y. Nakazato**, *SDSS+JWST Census of Stellar and Nebular Dust Attenuation at $z \sim 0$–7: Mass Dependence and Redshift Evolution*, 2026, submitted to ApJ, arXiv:2605.23049

26. K. Takechi, et al. (incl.**Y. Nakazato**), *DREAMS. JWST Spectroscopy of a $z = 8.3$ Galaxy with an ALMA Dust Continuum Detection: Early Dust, Very High Tdust, and a Multi-wavelength [O III] Ratio Discrepancy*, 2026, arXiv:2605.14922

¶25. C. Blanco-Prieto, et al. (incl.**Y. Nakazato**), *RIOJA. A clumpy, dust-rich merger at $z = 7.13$: JWST and ALMA characterization of A1689-zD1*, submitted to A& A

24. A. Faisst, L. Liu, Y. Dubois, **Y. Nakazato**, et al. (additional 53 authors), *The ALPINE-CRISTAL-JWST Survey: The Fast Metal Enrichment of Massive Galaxies at $z \sim 5$*, 2026, accepted to ApJ, arXiv:2510.16106

*23. D. Ceverino, **Y. Nakazato**, N. Yoshida et al., *Accelerated size evolution in the FirstLight simulations from $z = 14$ to $z = 5$*, 2026, arXiv:2603.05045

22. K. Mawatari, et al. (incl.**Y. Nakazato**), *RIOJA. Merger-induced Clumps in a Galaxy at Redshift 6.81 Revealed by JWST*, ApJ, 998, 119, 2026

21. Y. Harikane, P. Pérez-González, J. Álvarez-Márquez, M. Ouchi, **Y. Nakazato**, et al. (additional 6 authors), *A UV-Luminous Galaxy at $z = 11$ with Surprisingly Weak Star Formation Activity*, 2026, **submitted to Nature**, arXiv:2601.21833

20. Y. Ren, et al. (incl.**Y. Nakazato**), *RIOJA. Young Starburst and Ionized Gas Outflows in a $z = 7.212$ Galaxy Uncovered by JWST NIRCam and NIRSpec Observations*, 2025, MNRAS, 544, 4722, 2025 - *article (link)*

¶†19. T. Kiyota, M. Ouchi, Y. Xu, **Y. Nakazato**, et al. (additional 13 authors), *Comprehensive JWST+ALMA Study on the Extended Lyα Emitters, Himiko and CR7 at $z \sim 7$: Blue Major Merger Systems in Stark Contrast to Submillimeter Galaxies*, ApJ, 995, 150, 2025

†18. A. Crespo Gomez, et al. (incl.**Y. Nakazato**), *RIOJA. Dusty outflows and density-complex ISM in the N-rich lensed galaxy RXCJ2248-ID at $z = 6.1$*, 2025, arXiv:2511.14658

17. H. Umehata, Y. Tamura, Y. Fudamoto, **Y. Nakazato**, et al. (additional 16 authors), *ELPIS: Accelerated metal and dust enrichment in a proto-cluster core at $z \approx 8$*, ApJL, 993, L50, 2025

†16. R. Marques-Chaves, et al. (incl.**Y. Nakazato**), *Extremely UV-bright starbursts at the end of cosmic reionization*, 2025, arXiv:2510.12411

∗¶†15. Y. Fudamoto, **Y. Nakazato**, D. Ceverino, et al. (additional 24 authors), *Early massive galaxy formation in the core of a galaxy protocluster 650 million years after the Big Bang*, 2025, **under review in Nature Astronomy**, arXiv:2510.11770

†14. M. Usui, et al. (incl.**Y. Nakazato**), *RIOJA. JWST and ALMA Unveil the Inhomogeneous and Complex Interstellar Medium Structure in a Star-forming Galaxy at $z = 6.81$*, 2025, ApJL, 991, 2, L38, 2025

¶13. N. E. P. Lines, R. A. A. Bowler, N. J. Adams, R. Fisher, R. G. Varadaraj, **Y. Nakazato** et al. (additional 25 authors), *JWST PRIMER:A lack of outshining in four normal $z = 4-6$ galaxies from the ALMA-CRISTAL Survey*, MNRAS, 539, 3, 2025

12. Y. Sugahara, et al. (incl.**Y. Nakazato**), *RIOJA. Complex Dusty Starburst in a Major Merger B14-65666 at $z = 7.15$*, ApJ, 981, 135, 2025

11. N. Ishii et al. (incl.**Y. Nakazato**), *Detection of the [OI] 63μm emission line from the $z = 6.04$ quasar J2054-0005*, PASJ, 77, 139, 2025

†10. Y. Harikane, A. K. Inoue, R. Ellis, M. Ouchi, **Y. Nakazato**, et al. (additional 31 authors), *JWST, ALMA, and Keck Spectroscopic Constraints on the UV Luminosity Functions at $z \sim 7-14$: Clumpiness and Compactness of the Brightest Galaxies in the Early Universe*, ApJ, 980, 138, 2025

∗¶†9. T. S. Tanaka, J. D. Silverman, **Y. Nakazato**, et al. (additional 40 authors), *Crimson Behemoth: a Massive Clumpy Structure Hosting a Dusty AGN at $z = 4.91$*, PASJ, psae091, 2024

∗8. D. Ceverino, **Y. Nakazato**, N. Yoshida, R.S. Klessen, S.C.O. Glover, *Redshift-dependent galaxy formation efficiency at $z = 5-13$ in the FirstLight Simulations*, A & A 489, A244. 2024,

7. C. Williams, et al. (incl.**Y. Nakazato**), *The Supersonic Project: Lighting up the faint end of the JWST UV luminosity function*, ApJL, 960, 2, 16, 2024
   - *press released (link)*

∗6. D. Tsuna, **Y. Nakazato**, T. Hartwig, *A Photon Burst Clears the Earliest Dusty Galaxies: Modeling Dust in High-redshift Galaxies from ALMA to JWST*, MNRAS, 526, 4, 2023

5. W. Lake et al.(incl.**Y. Nakazato**), *The Supersonic Project: Star Formation in Early Star Clusters without Dark Matter*, ApJL, 956, 1, 2023

¶†4. T. Hashimoto et al.(incl.**Y. Nakazato**), *Reionization and the ISM/Stellar Origins with JWST and ALMA (RIOJA):The core of the highest redshift galaxy overdensity at $z = 7.88$ confirmed by NIRSpec/JWST*, ApJL, 955, 8, 2023
   - *press released (link)*

3. R. Ura, et al.(incl.**Y. Nakazato**), *Detections of [CII] 158 μm and [OIII] 88 μm in a Local Lyman Continuum Emitter, Mrk 54, and Its Implications to High-redshift ALMA Studies*, ApJ, 948, 1, 2023

2. C. Williams, et al.(incl.**Y. Nakazato**), *The Supersonic Project: The eccentricity and rotational support of SIGOs and DM GHOSts*, ApJ, 945, 1, 2023

1. W. Lake, et al.(incl.**Y. Nakazato**), *The Supersonic Project: The Early Evolutionary Path of SIGOs*, ApJ, 943, 2, 2023

Co-authored Proceedings & Other Reports (Refereed)

1. T. Hashimoto, et al.(incl.**Y. Nakazato**), *Far-infrared line properties of Green Pea galaxies, the best local analogues of galaxies in the epoch of reionization era*, PRIMA General Observer Science Book Volume 2